



THE NATIONAL INSTITUTE OF ENGINEERING

Mysuru – 570008

Department of Electronics and Communication Engineering

LINEAR INTEGRATED CIRCUITS LAB REPORT

Submitted by:

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Submitted to:

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C3 TEAM - 3

Design Question:

Design precision rectifier circuits as per the lab sheet:

- Half wave rectifier → Hardware
- Full wave rectifier → Simulation

Given:

- $V_{CC} = \pm 12V$

Also, change the direction of diodes and observe the output waveforms.

AIM:

To design and simulate half-wave and full-wave precision rectifier circuits using an operational amplifier with a $\pm 12V$ supply, and to observe the effect of changing diode direction on the output waveforms.

Components Required :

- Operational Amplifier ($\mu A741$)
- Diodes (1N4148)
- Resistors (as per design requirement)
- DC Power Supply ($\pm 12V$)
- Input Signal Source (Sine wave: 100 m Vp-p, 1 kHz)
- LTspice Software

THEORY :

A rectifier is a circuit that converts an AC signal into a DC signal. In conventional rectifiers using diodes, a major limitation is the **cut-in voltage** of the diode (approximately 0.7 V for silicon diodes). Because of this, small input signals (in millivolts) cannot be properly rectified.

To overcome this limitation, **precision rectifiers** are used. A precision rectifier uses an **operational amplifier (op-amp) with a diode** in its feedback path. Due to the high open-loop gain of the op-amp, the effect of the diode's threshold voltage is minimized, making the diode behave like an **ideal diode** (zero cut-in voltage). This combination is also called a **super diode**.

Half-Wave Precision Rectifier

In a half-wave precision rectifier:

- The circuit allows only one half (positive or negative) of the input signal to appear at the output.
- During one half-cycle, the diode conducts and the op-amp operates in a closed-loop configuration.
- During the other half-cycle, the diode is reverse-biased and output becomes zero.

Thus, the output is an accurate rectified version of the input without voltage drop.

Full-Wave Precision Rectifier

In a full-wave precision rectifier:

- Both positive and negative halves of the input signal are converted into a single polarity (usually positive).
- It uses multiple diodes and op-amps to process both halves of the input waveform.
- The output is a **full-wave rectified signal**, maintaining the same amplitude as the input.

Effect of Diode Direction

- Changing the direction of the diode changes which half-cycle is allowed at the output.
- This results in either **positive half-wave**, **negative half-wave**, or inversion of the waveform.

Advantages of Precision Rectifier

- Can rectify **very small signals (mV range)**
- Eliminates diode threshold voltage error
- High accuracy
- Suitable for **high-frequency applications**

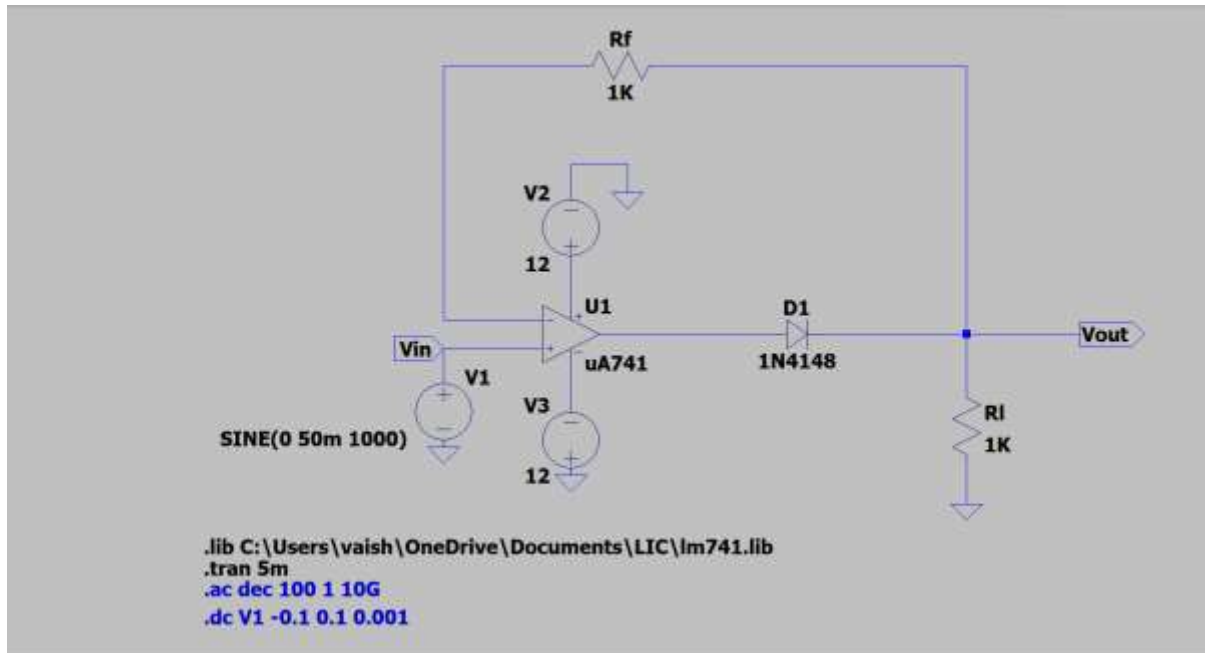
PROCEDURE

1. Open **LTspice** and create a new schematic.
2. Place the required components:
 - μ A741 Op-Amp
 - Diodes
 - Resistors
 - Voltage sources (for $\pm 12\text{V}$ supply and input signal)
3. Configure the **power supply**:
 - Set +12V to +VCC
 - Set -12V to -VCC
4. Apply the **input signal**:
 - Sine wave
 - Amplitude = 100 mVpp
 - Frequency = 1 kHz
5. Connect the circuit as per the **half-wave precision rectifier diagram**.
6. Run the simulation and observe:
 - Input waveform
 - Output waveform
7. Change the **direction of the diode** and repeat the simulation.
8. Note the change in output waveform for both cases.
9. Modify the circuit to implement the **full-wave precision rectifier**.
10. Run the simulation and observe the full-wave rectified output.
11. Change resistor values (if required) to obtain different gain values and observe the output.
12. Plot and record all **input and output waveforms**.

1 Non-Inverting Half Wave Precision Rectifier :

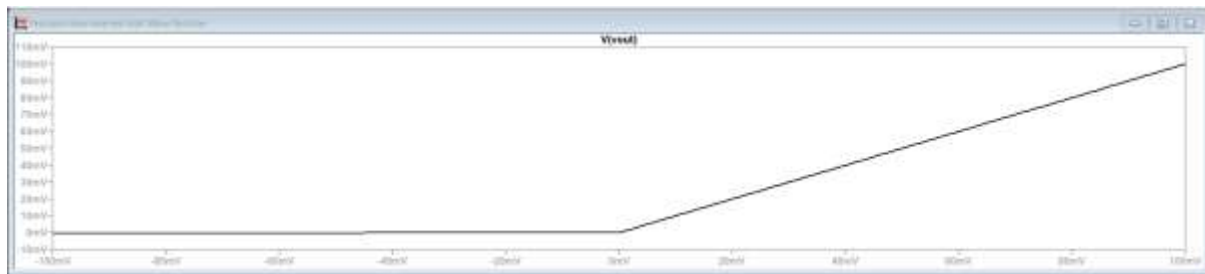
CASE 1: When the Diode is in Forward Bias

CIRCUIT DIAGRAM:

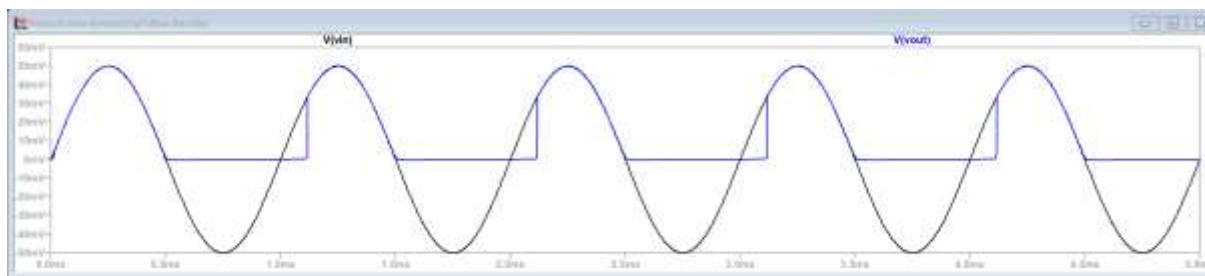


ANALYSIS :

1 DC SWEEP (TRANSFER CHARACTERISTICS) :



2 TRANSIENT :

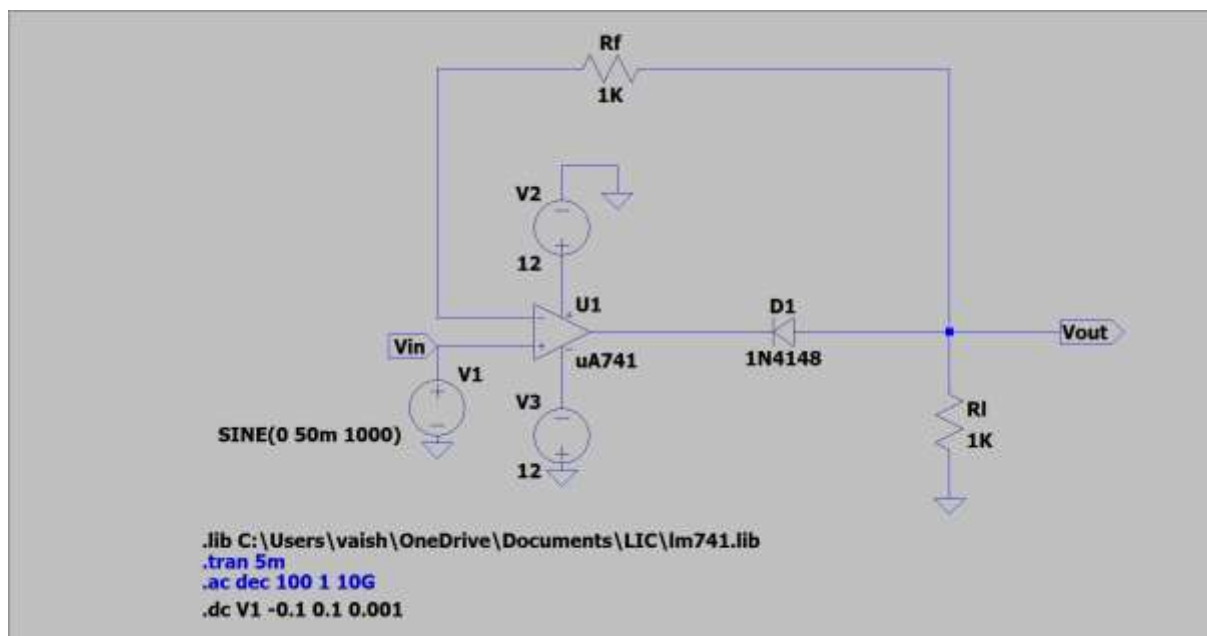


OBSERVATIONS:

When the sine wave crosses 0V and starts going positive again, the op-amp has to "wake up" and get back to the positive side to turn the diode back on. However, it is starting from -12V. The op-amp has to travel all the way from -12V up to +0.7V (the diode's turn-on voltage) before Vout can do anything. Because the mu A741 is an older op-amp, it has a slow Slew Rate.

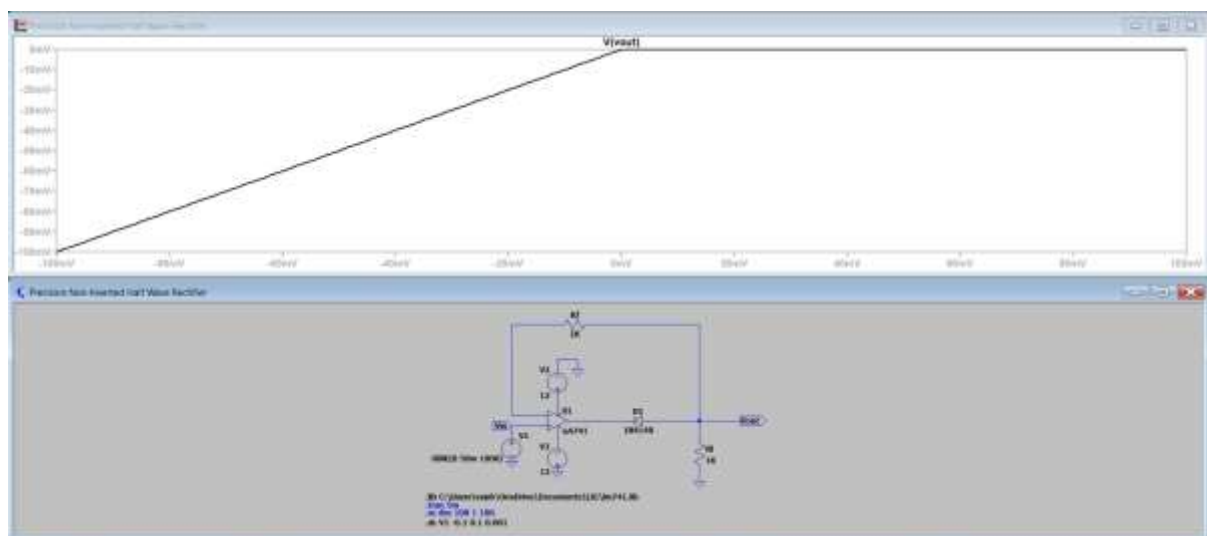
CASE 2: When the Diode is in Reverse Bias

CIRCUIT DIAGRAM:

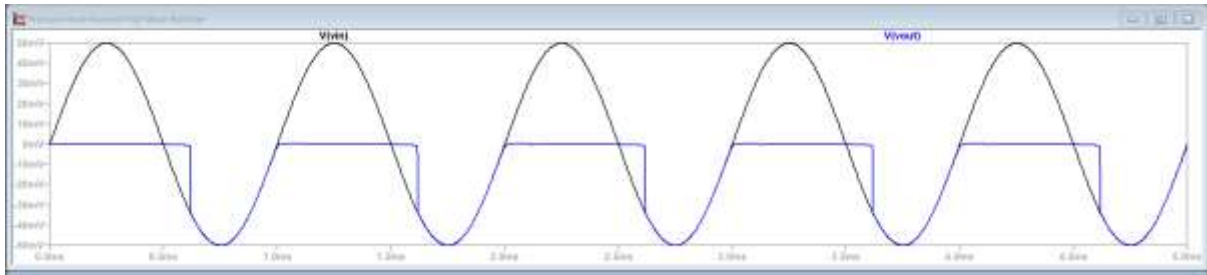


ANALYSIS :

1 DC SWEEP (TRANSFER CHARACTERISTICS) :



2 TRANSIENT :



OBSERVATIONS :

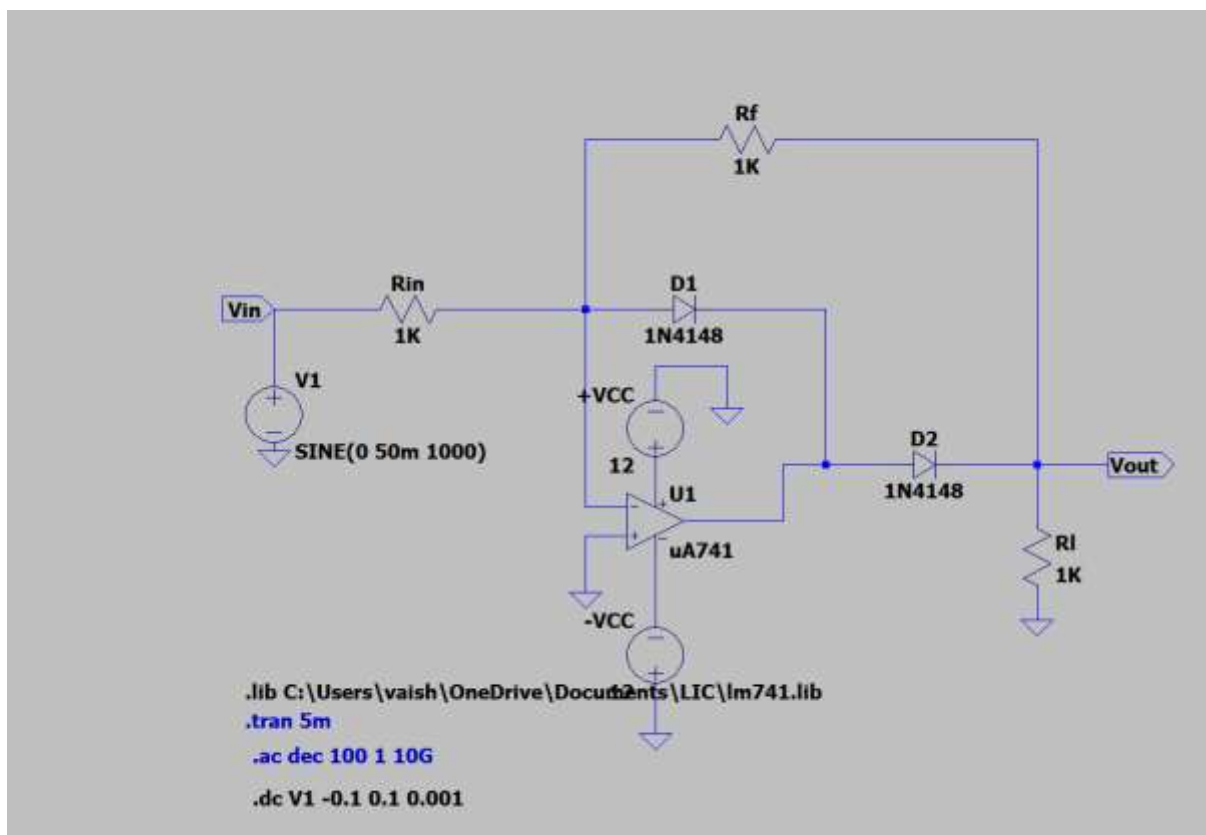
When input V_{in} goes negative :

The op-amp's non-inverting (+) terminal is lower than the inverting (-) terminal. The op-amp output responds by swinging negative. Since the diode's cathode (the side with the line) is connected to the op-amp and the anode is connected to V_{out} , a negative swing at the output forward-biases the diode. This "closes the gate," completing the feedback loop. The op-amp now forces V_{out} to perfectly match V_{in} to keep the differential input at zero

2 Inverting Half Wave Precision Rectifier :

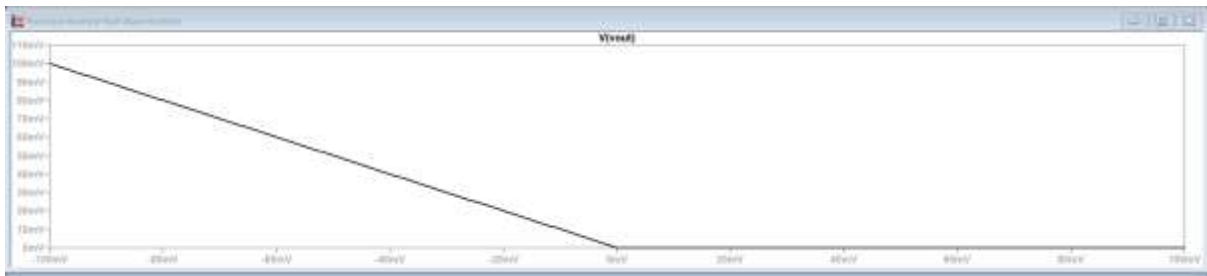
CASE 1: D1 cathode connects to op-amp output; D2 anode connects to op-amp output (Produces Positive Half-Cycles)

CIRCUIT DIAGRAM:

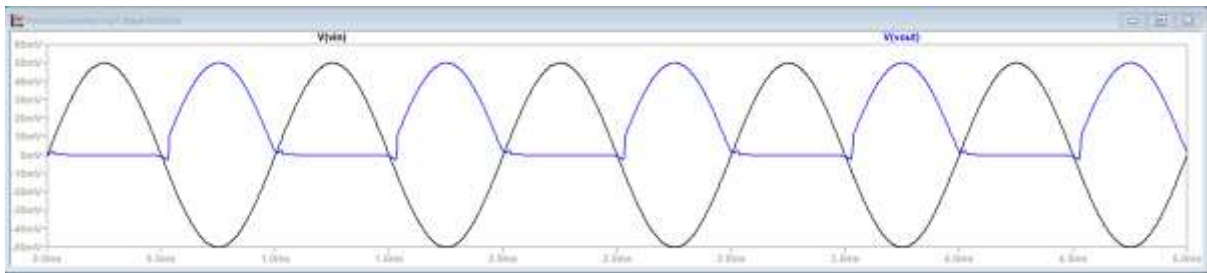


ANALYSIS :

1 DC SWEEP (TRANSFER CHARACTERISTICS) :

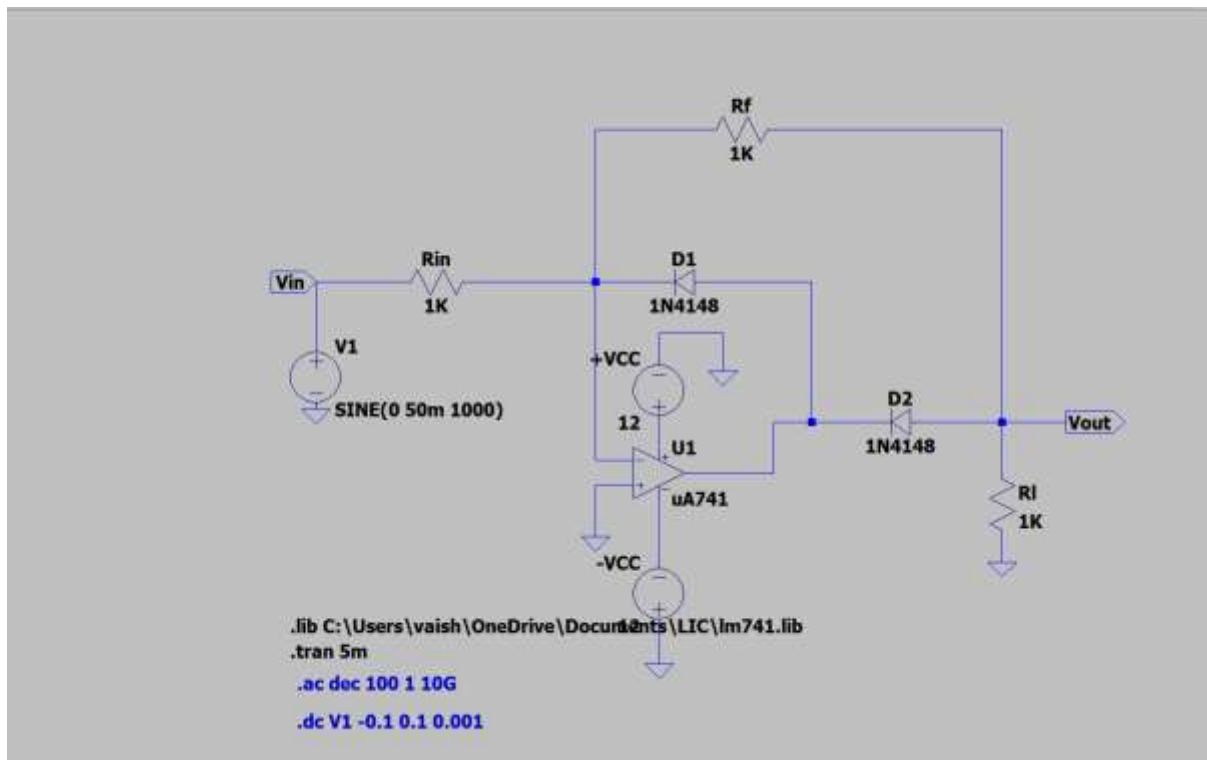


2 TRANSIENT :



CASE 2: Reversed Diodes Configuration (Produces Negative Half-Cycles)

CIRCUIT DIAGRAM

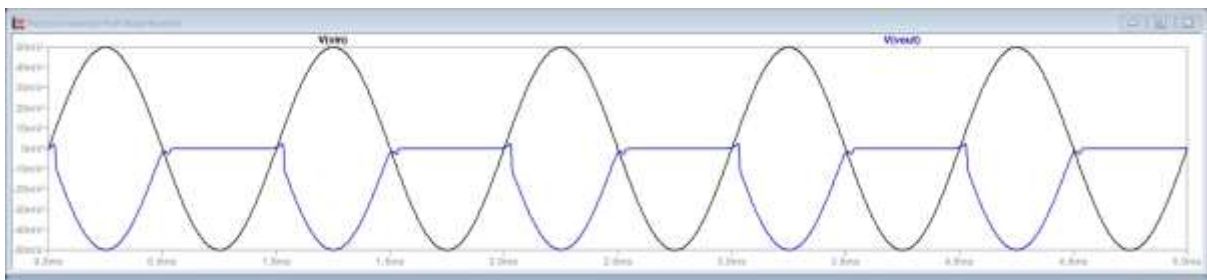


ANALYSIS :

1 DC SWEEP (TRANSFER CHARACTERISTICS) :



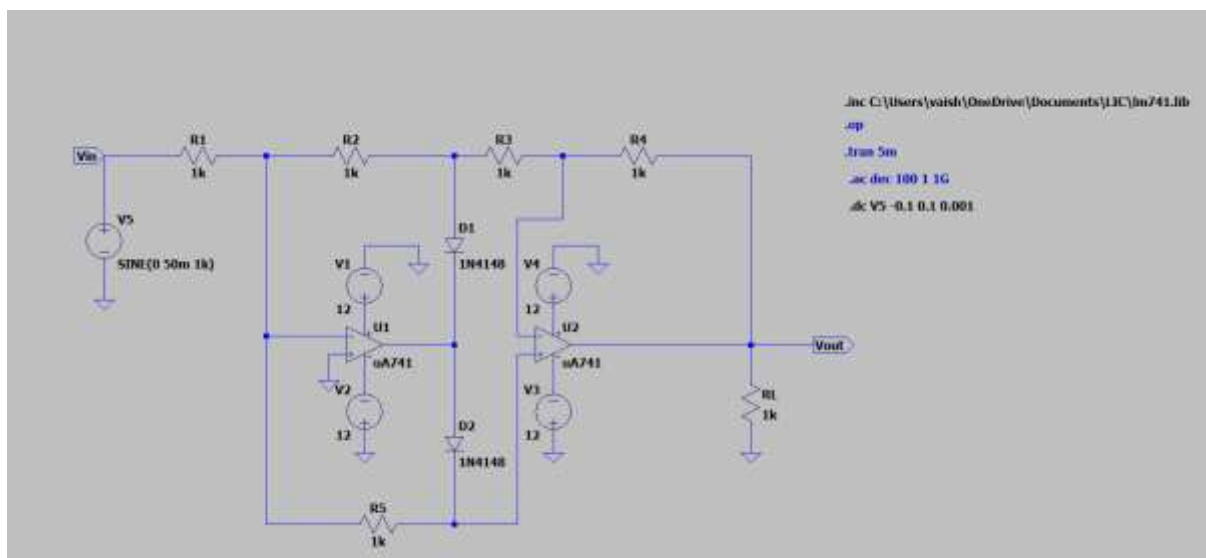
2 TRANSIENT :



3 Full Wave Precision Rectifier :

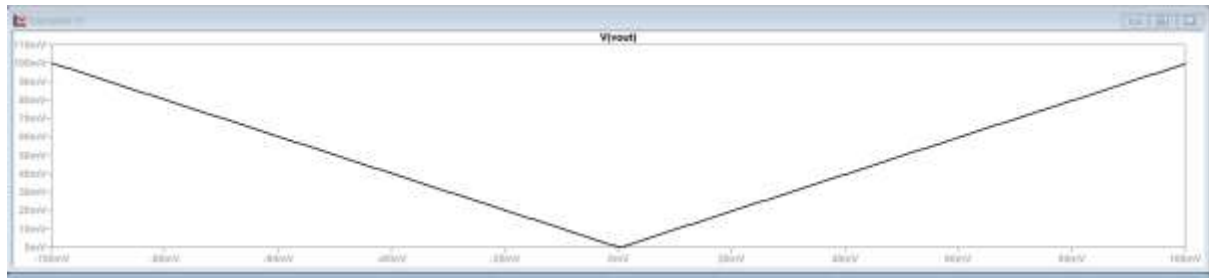
CASE 1: D1 and D2 both pointing Upwards in the schematic. D1 anode connected to U1 output; D2 cathode connected to U1 output (Produces Positive Full-Wave)

CIRCUIT DIAGRAM:

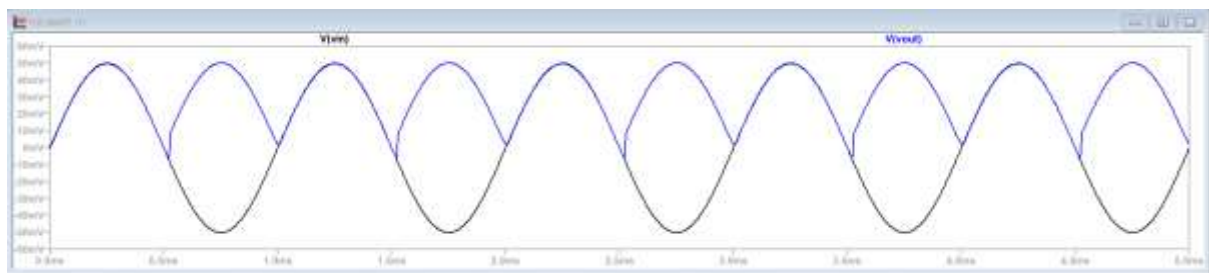


ANALYSIS :

1 DC SWEEP (TRANSFER CHARACTERISTICS) :

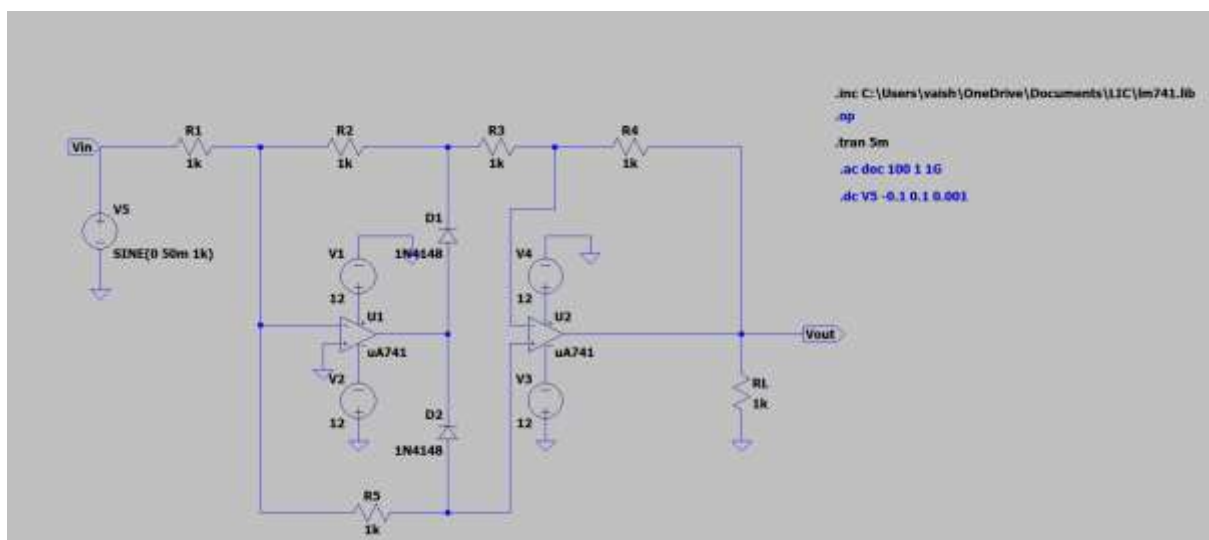


2 TRANSIENT :



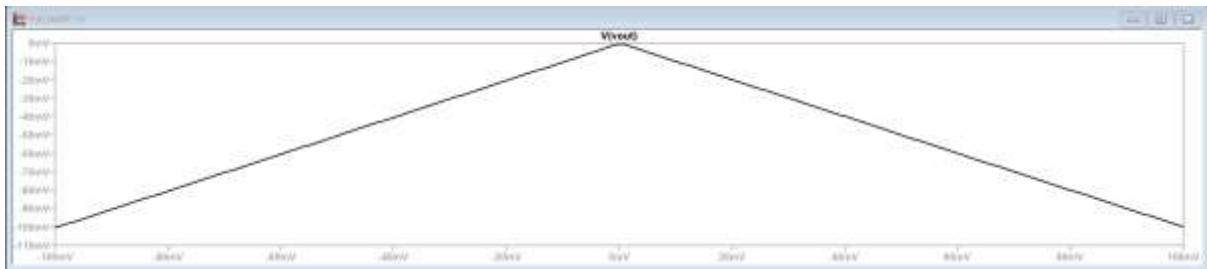
CASE 2: Reversed Diodes (Produces Negative Full-Wave)

CIRCUIT DIAGRAM:

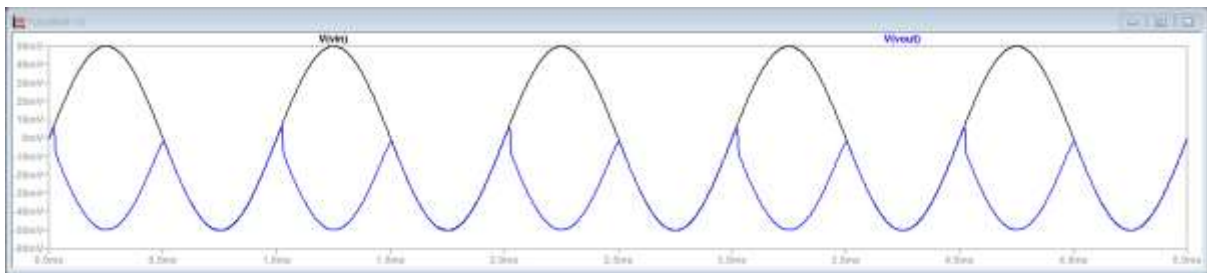


ANALYSIS :

1 DC SWEEP (TRANSFER CHARACTERISTICS) :



2 TRANSIENT :



RESULTS :

The precision rectifier circuits were designed and simulated successfully using LTspice with a $\pm 12\text{V}$ supply.

For the **non-inverting half wave precision rectifier**, the output was observed only during one half cycle of the input signal, while the other half was completely blocked. When the direction of the diode was reversed, the output waveform also reversed, allowing the opposite half cycle to appear at the output.

The **inverting half wave rectifier** also showed similar rectification behaviour, but with phase inversion. The output followed the input during the conducting half cycle with opposite polarity.

For the **full wave precision rectifier**, both positive and negative halves of the input signal were rectified and appeared as a single polarity output. The output waveform maintained the same amplitude as the input, showing that the effect of diode threshold voltage was eliminated.

From the **DC sweep analysis**, the transfer characteristics confirmed that the circuit behaves like an ideal rectifier, where output follows input only in the conducting region and remains zero otherwise.

INFERENCE :

From this experiment, it is clear that precision rectifiers overcome the limitations of conventional diode rectifiers, especially for low amplitude signals. Even for an input of 100 mV, the circuit was able to rectify properly without any noticeable voltage drop.

The op-amp plays a major role by compensating for the diode forward voltage, making the diode behave like an ideal one.

It was also observed that the direction of the diode directly affects which half of the signal is passed to the output. This helped in understanding how rectification can be controlled.

In DC sweep, the transfer curve clearly showed the regions where the diode conducts and where it blocks, which matches the theoretical behaviour.

Overall, the experiment helped in understanding the working of precision rectifiers, their advantages over normal rectifiers, and their importance in low-voltage signal processing applications.